Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	V
Subject Code & Name	UCS2612 & Machine Learning Algorithms Laboratory		
Academic year	2025-2026 (Even)	Batch:2023-2027	**Due date: 23/12/25**

Experiment 1: Working with Python packages – Numpy, Scipy, Scikit-Learn, Matplotlib

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Aim:
To understand and explore Python packages like Numpy, Scikit-learn, and Matplotlib on public datasets and identify ML tasks, feature selection techniques, and suitable algorithms.

Libraries used:

- Matplotlib.pyplot
- Seaborn
- Numpy
- Pandas

Mathematical and Theoretical Explanation of the Performed Tasks:

- **Primary Objective:** Exploratory Data Analysis (EDA) is carried out to gain an initial understanding of the dataset by analyzing feature distributions, identifying anomalies, examining class proportions, and studying relationships and dependencies among variables.
- **Statistical Measures**
  - Arithmetic Mean:
  $$\mu = \frac{1}{n}\sum_{i=1}^{n} x_i$$
  - Sample Variance:
  $$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \mu)^2$$
  - Quartiles (Median, Q_1, Q_3) are used to describe central tendency and dispersion, forming the basis of boxplot visualization.
- **Histogram Analysis**
  - Histograms represent the empirical frequency distribution of numerical features, helping to visualize data spread, skewness, and dominant values.

– Kernel Density Estimation (KDE) is employed to obtain a smooth approximation of the underlying probability density function:

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right)$$

where $K(\cdot)$ denotes the kernel function and h represents the smoothing parameter.

- **Boxplot Representation**
  - Boxplots summarize data using the median and quartiles, while the Interquartile Range (IQR) is defined as:

    $$IQR = Q_3 - Q_1$$
  - Data points lying beyond $1.5 \times IQR$ from the quartiles are treated as potential outliers.
- **Scatter and Pairwise Visualizations**
  - Scatter plots are used to explore possible linear or nonlinear associations between feature pairs.
  - Pairplots extend this analysis by displaying marginal distributions along the diagonal using histograms or KDE plots.
- **Correlation Analysis**
  - Linear relationships between variables are quantified using the Pearson correlation coefficient:

    $$r_{XY} = \frac{\mathrm{cov}(X, Y)}{\sigma_X \sigma_Y}$$
  - The covariance between two variables is given by:

    $$\mathrm{cov}(X, Y) = \frac{1}{n-1} \sum (X_i - \bar{X})(Y_i - \bar{Y})$$
  - Correlation heatmaps visually convey the strength and direction of relationships, with values ranging from -1 to $+1$.
- **Categorical Data Visualization**
  - Count plots are used to display the frequency of categorical variables and analyze class-wise distributions when combined with class labels.
- **Interpretational Objectives**
  - The analysis aims to detect skewed distributions, multiple modes, outliers, correlated features, and class imbalance, which influence preprocessing decisions and model selection.

Results and Discussions:

The dataset used here is the Diabetes dataset, which considers various factors such as gender, age, hypertension and other valid health factors to predict and classify classes as Diabetic and Non-Diabetic.

Results and Discussions:

0.1 Loan Amount Prediction

...	person_age	person_gender	person_education	person_income	person_emp_exp	person_home_ownership	loan_amnt	loan_in
0	22.0	female	Master	71948.0	0	RENT	35000.0	PERSC
1	21.0	female	High School	12282.0	0	OWN	1000.0	EDUCA
2	25.0	female	High School	12438.0	3	MORTGAGE	5500.0	MED
3	23.0	female	Bachelor	79753.0	0	RENT	35000.0	MED
4	24.0	male	Master	66135.0	1	RENT	35000.0	MED
...	...	...	...	...	...	...	...	
44995	27.0	male	Associate	47971.0	6	RENT	15000.0	MED
44996	37.0	female	Associate	65800.0	17	RENT	9000.0	HOMEIMPROVEM
44997	33.0	male	Associate	56942.0	7	RENT	2771.0	DEBTCONSOLIDA
44998	29.0	male	Bachelor	33164.0	4	RENT	12000.0	EDUCA
44999	24.0	male	High School	51609.0	1	RENT	6665.0	DEBTCONSOLIDA

45000 rows × 14 columns

Figure 1: Dataset Columns

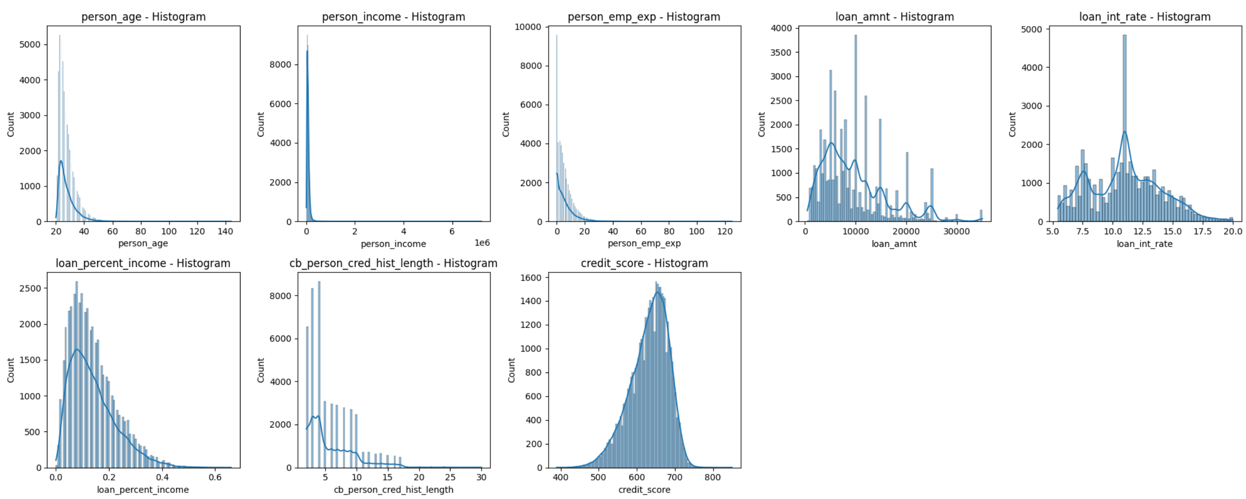

Figure 2: Loan Amount Distribution(histogram plot)

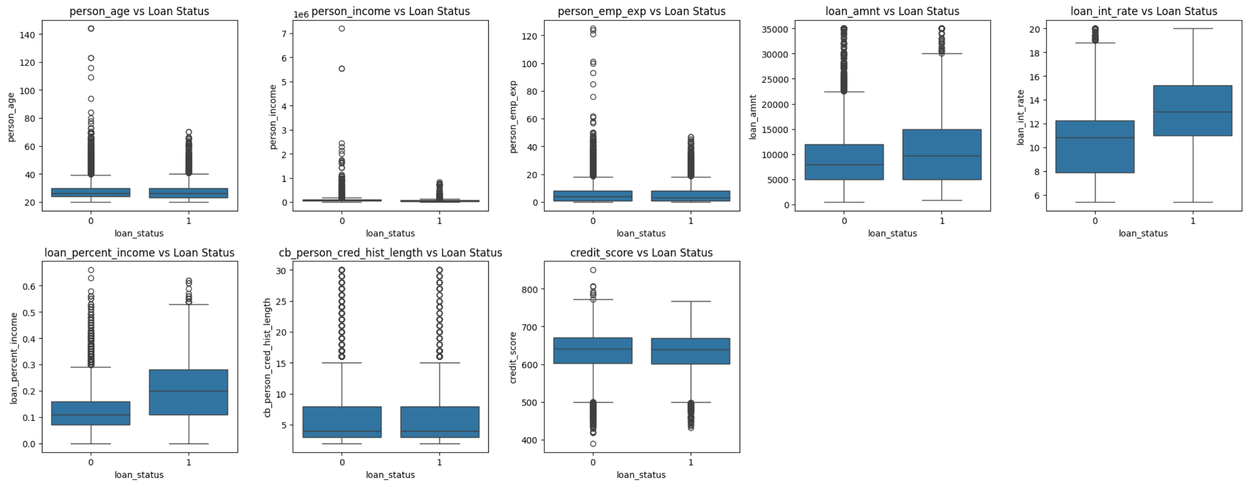

Figure 3: Boxplot Distribution

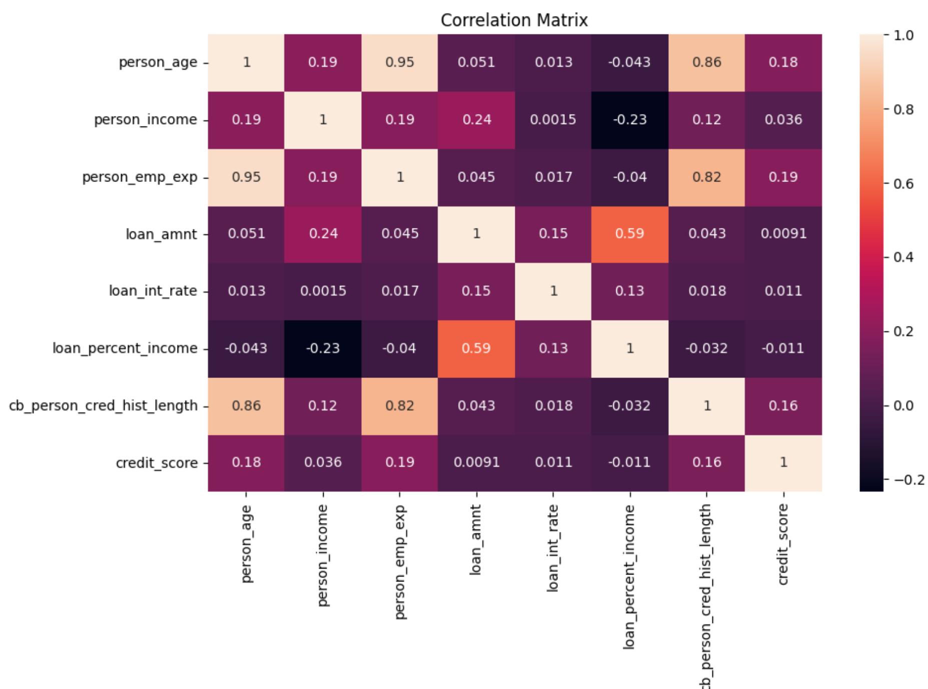

Figure 4: Correlation Matrix

0.2 Predicting Diabetes

	gender	age	hypertension	heart_disease	smoking_history	bmi	HbA1c_level	blood_glucose_level	diabetes
0	Female	80.0	0	1	never	25.19	6.6	140	0
1	Female	54.0	0	0	No Info	27.32	6.6	80	0
2	Male	28.0	0	0	never	27.32	5.7	158	0
3	Female	36.0	0	0	current	23.45	5.0	155	0
4	Male	76.0	1	1	current	20.14	4.8	155	0
...	...	...	...	...	...	...	...	...	...
99995	Female	80.0	0	0	No Info	27.32	6.2	90	0
99996	Female	2.0	0	0	No Info	17.37	6.5	100	0
99997	Male	66.0	0	0	former	27.83	5.7	155	0
99998	Female	24.0	0	0	never	35.42	4.0	100	0
99999	Female	57.0	0	0	current	22.43	6.6	90	0

100000 rows × 9 columns

Figure 5: Dataset Columns

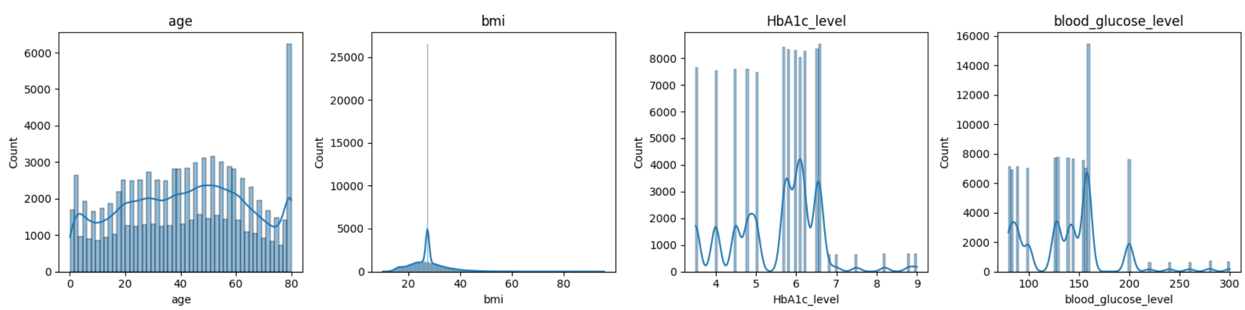

Figure 6: Histogram Distribution

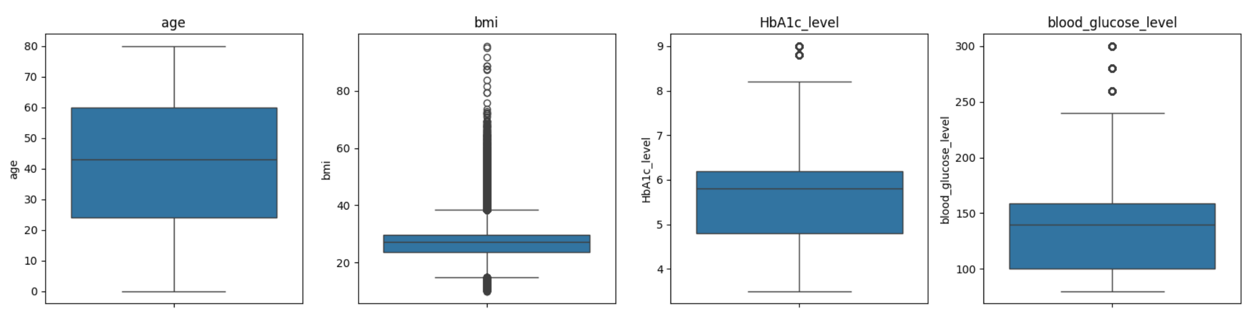

Figure 7: Boxplot distribution

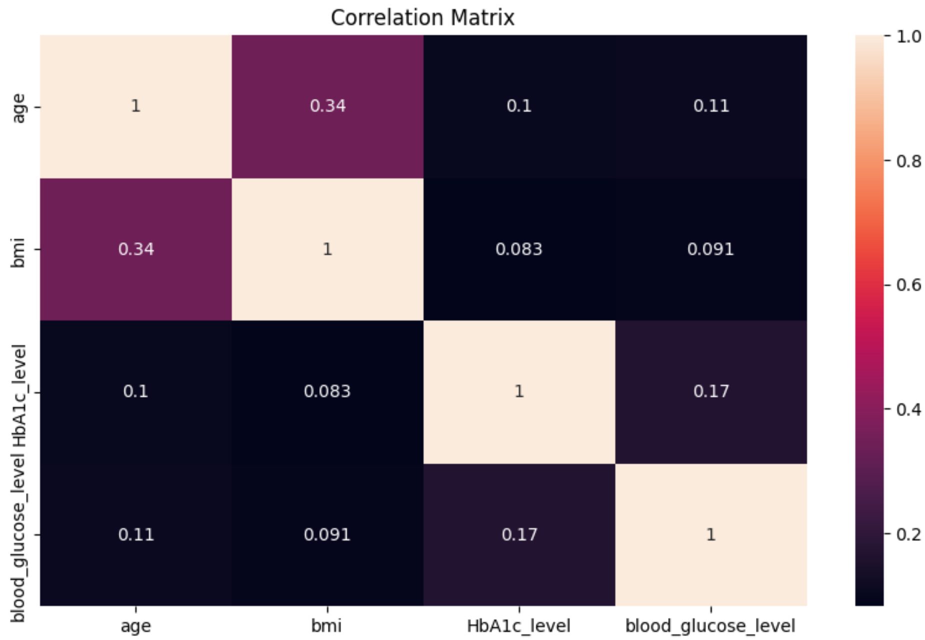

Figure 8: Correlation Matrix

0.3 Classification of Email Spam

	Unnamed: 0	label	text	label_num
0	605	ham	Subject: enron methanol ; meter # : 988291\r\n...	0
1	2349	ham	Subject: hpl nom for january 9 , 2001\r\n(see...	0
2	3624	ham	Subject: neon retreat\r\nho ho ho , we ' re ar...	0
3	4685	spam	Subject: photoshop , windows , office . cheap ...	1
4	2030	ham	Subject: re : indian springs\r\nthis deal is t...	0
...	...	...	...	...
5166	1518	ham	Subject: put the 10 on the ft\r\nthe transport...	0
5167	404	ham	Subject: 3 / 4 / 2000 and following noms\r\nhp...	0
5168	2933	ham	Subject: calpine daily gas nomination\r\n>\r\n...	0
5169	1409	ham	Subject: industrial worksheets for august 2000...	0
5170	4807	spam	Subject: important online banking alert\r\ndea...	1

Figure 9: dataset columns

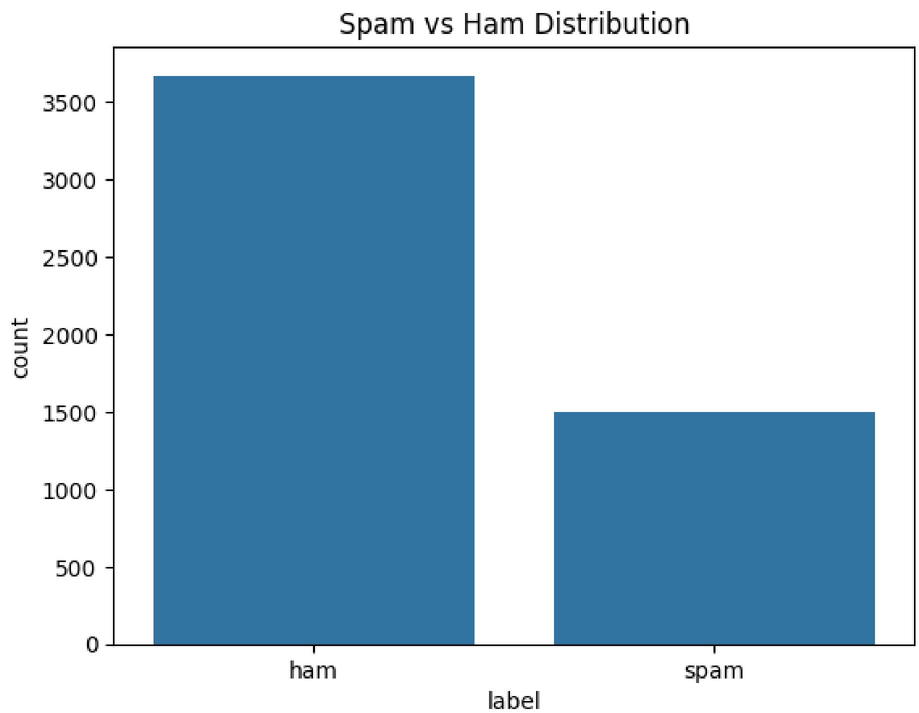

Figure 10: spam vs ham distribution

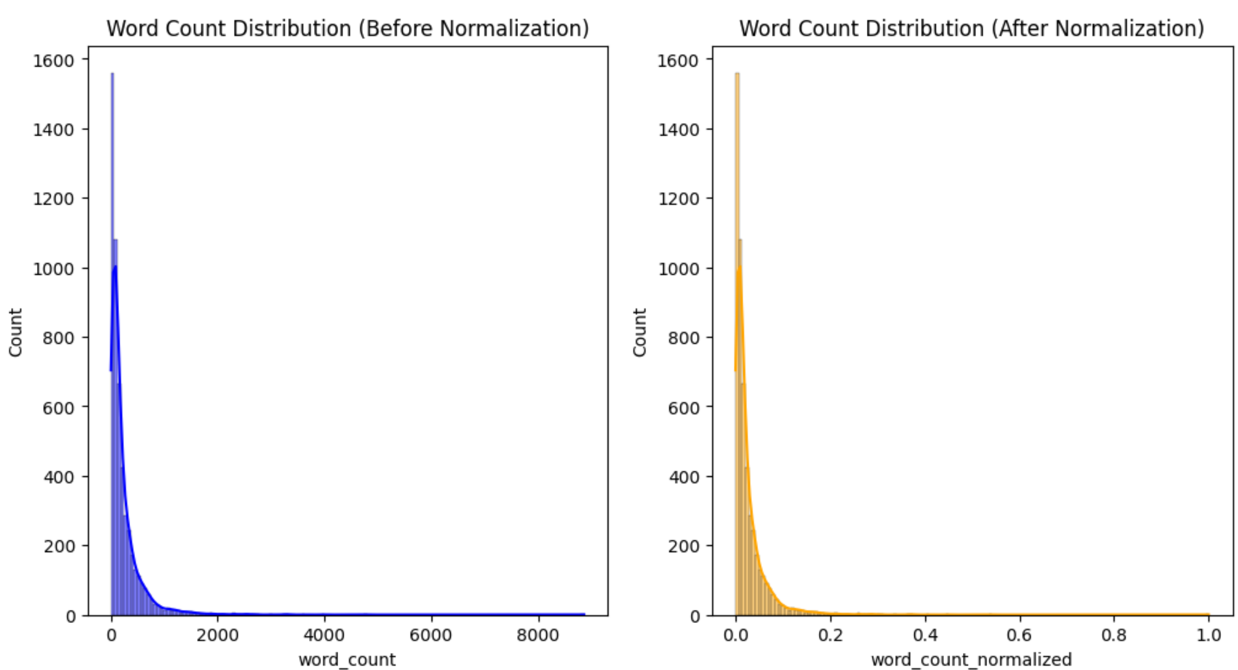

Figure 11: word count distribution(Normalisation)

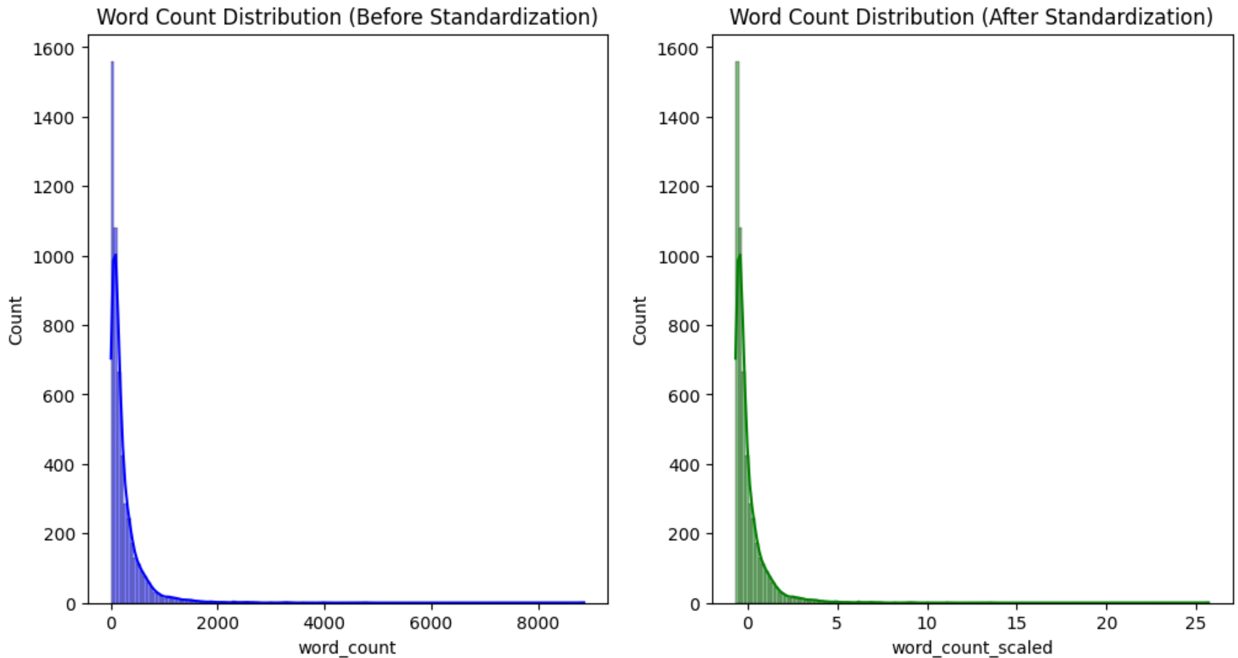

Figure 12: word count distribution(Standardisation)

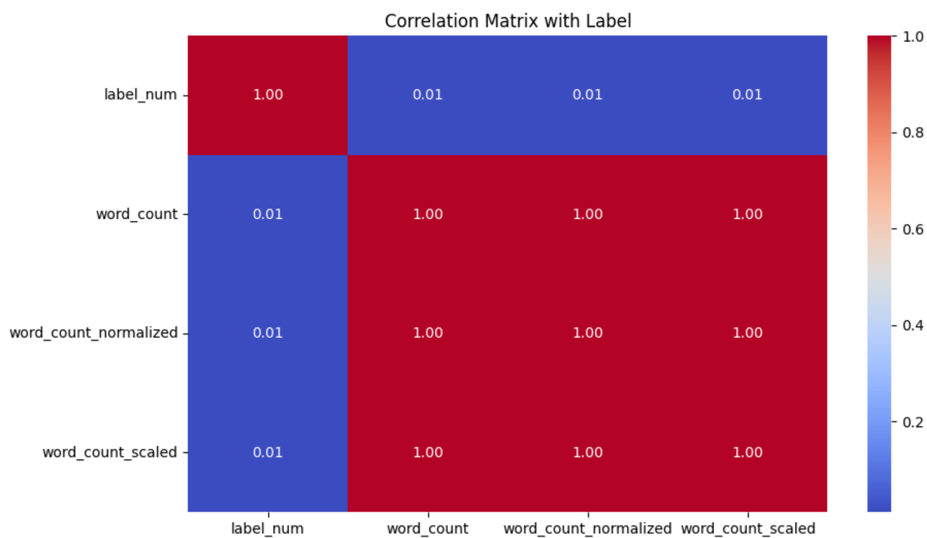

Figure 13: Correlation Matrix

0.4 Handwritten Character Recognition (MNIST)

Figure 14: sample images

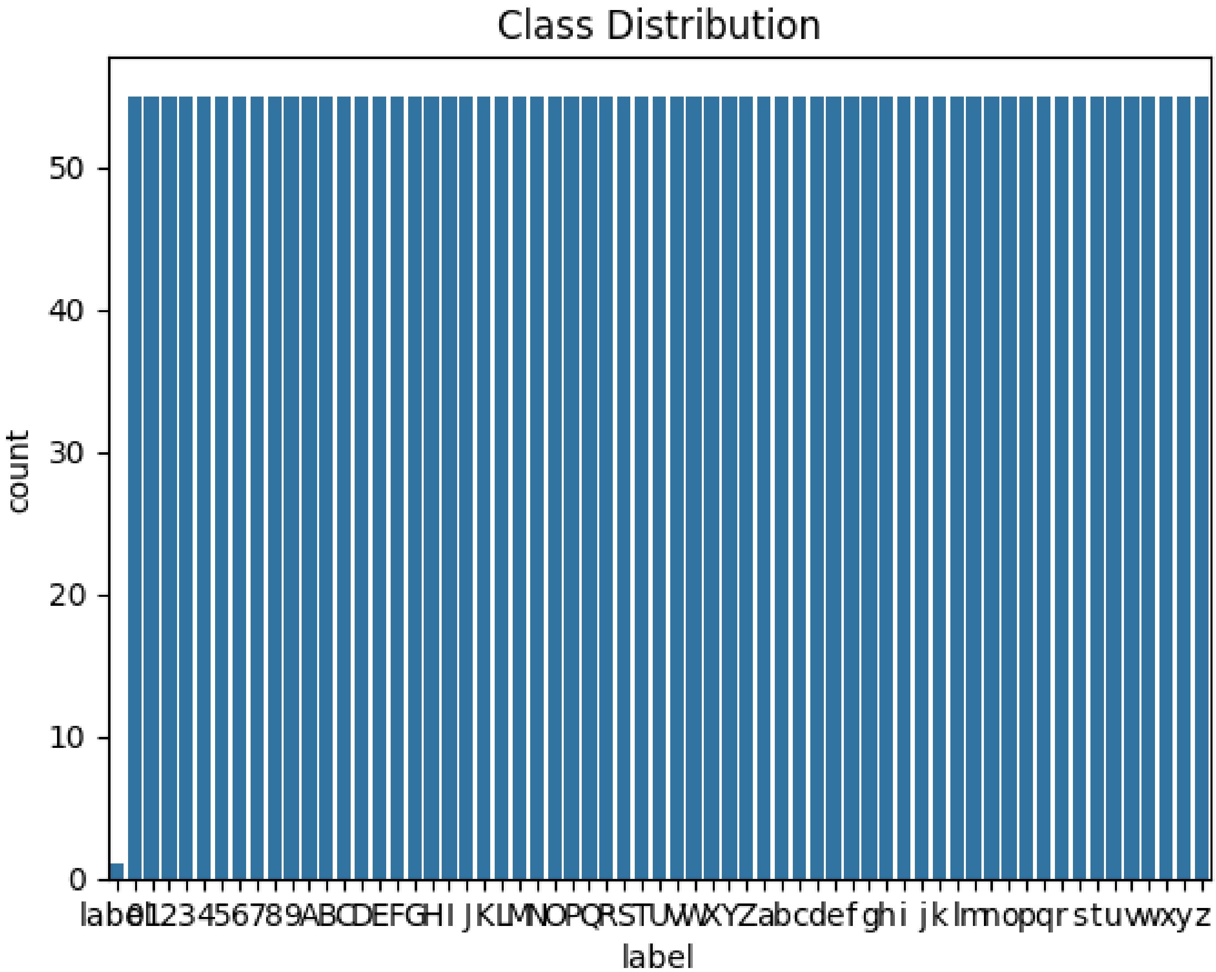

Figure 15: Digit Distribution

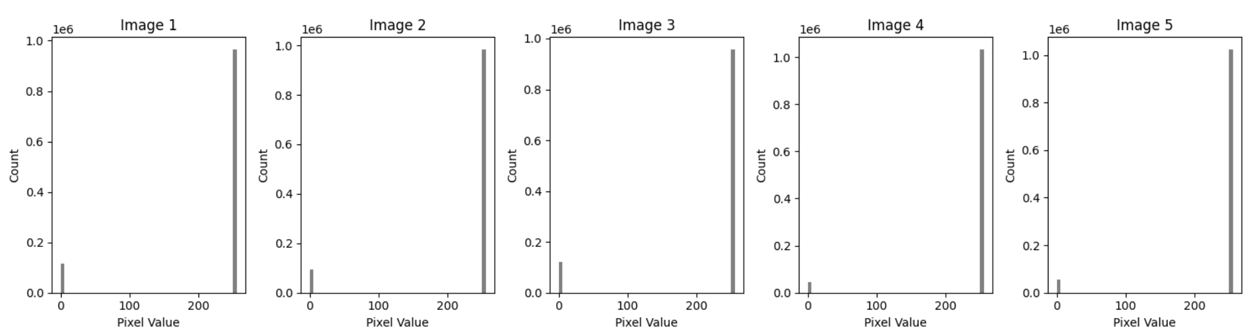

Figure 16: Pixel Intensity Analysis

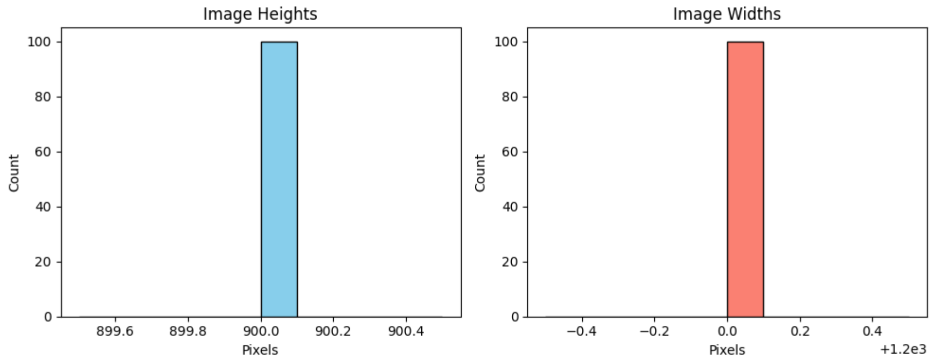

Figure 17: Height weight distribution

0.5 Iris Dataset

	sepal_length	sepal_width	petal_length	petal_width	species
0	NaN	NaN	NaN	NaN	Species
1	4.9	3.0	1.4	0.2	Iris-setosa
2	4.7	3.2	1.3	0.2	Iris-setosa
3	4.6	3.1	1.5	0.2	Iris-setosa
4	5.0	3.6	1.4	0.2	Iris-setosa

Figure 18: Dataset Columns

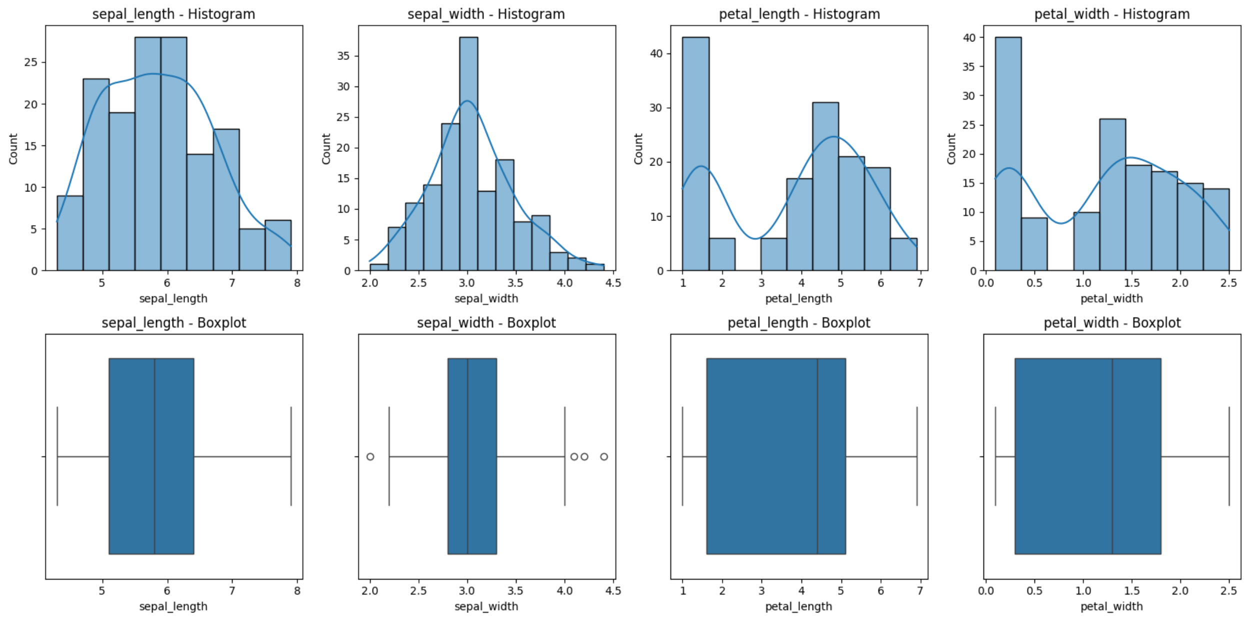

Figure 19: Histogram and Boxplot Distribution

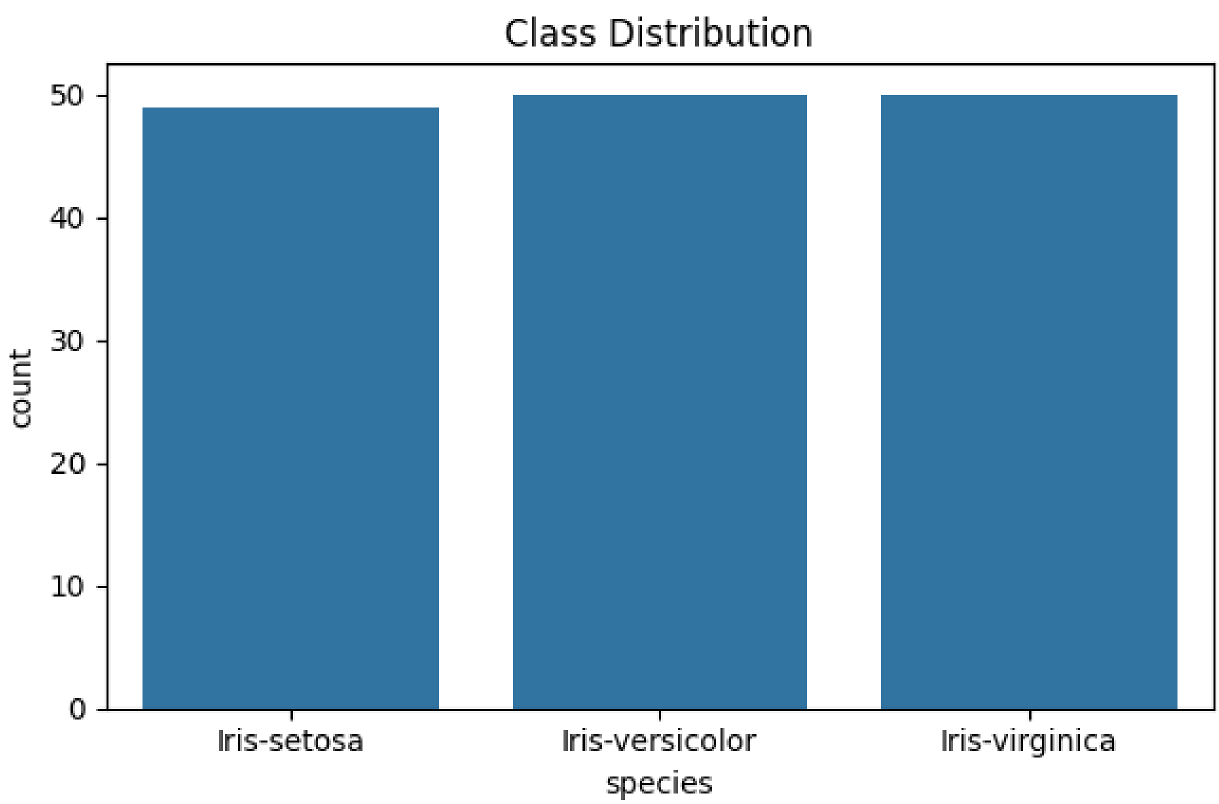

Figure 20: class distribution

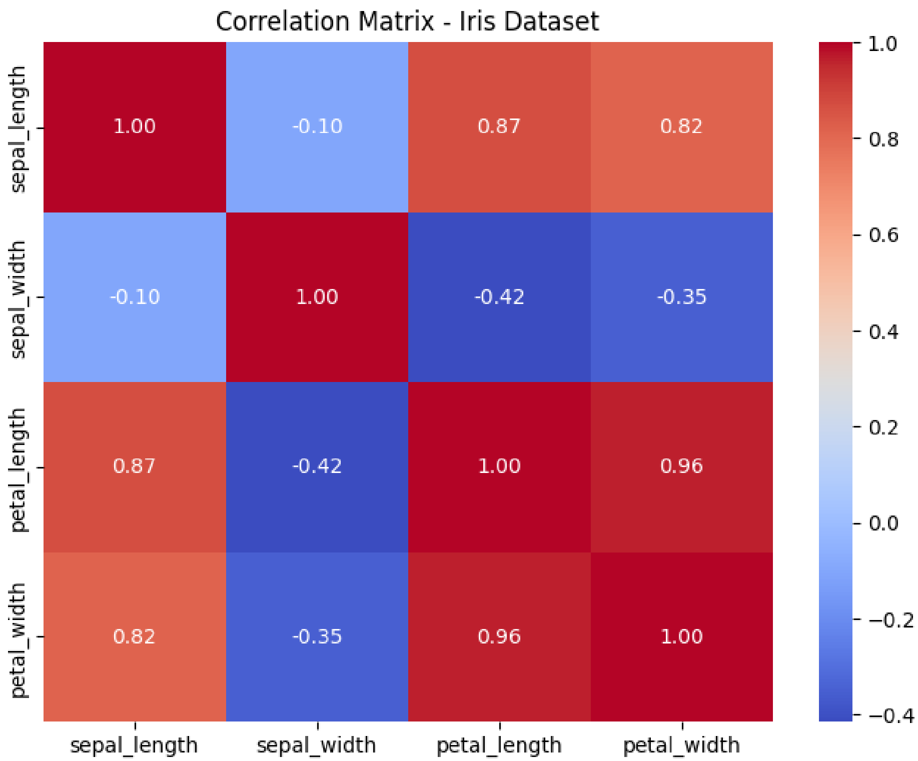

Figure 21: Correlation Matrix

Dataset	**Type of ML Task**	**Feature Selection Technique**	**Suitable ML Algorithm**
Iris Dataset	Multi-class Classification	Correlation Matrix / ANOVA F-value	k-Nearest Neighbors (k-NN), Decision Trees
Loan Amount Prediction	Regression	Recursive Feature Elimination (RFE) / Pearson Correlation	Linear Regression, Random Forest Regressor
Predicting Diabetes	Binary Classification	Chi-Square Test / SelectKBest	Logistic Regression, Support Vector Machine (SVM)
Classification of Email Spam	Binary Classification (NLP)	Information Gain / Chi-Square (on word vectors)	Naive Bayes, SVM
Handwritten Character Recognition / MNIST	Multi-class Image Classification	Principal Component Analysis (PCA)	Convolutional Neural Networks (CNN), SVM

Learning Practices:

- Interpret dataset structure: Learn to inspect shape, info, and missing values.
- Visualize distributions: Gain skills in plotting histograms, boxplots, and correlation heatmaps.
- Identify class balance: Understand how label distribution affects model performance.